

Article Title: “RoBERTa-GCN: A Novel Approach for Combating Fake News in Bangla Using Advanced Language Processing and Graph Convolutional Networks”

To: IEEE Access Editor

Re: Response to reviewers

Dear Editor,

Thank you for allowing a resubmission of our manuscript, with an opportunity to address the reviewers' comments. We are uploading (a) our point-by-point response to the comments (below) (response to reviewers), (b) an updated manuscript with yellow highlighting indicating changes, and (c) a clean updated manuscript without highlights (Main Manuscript).

Best regards,

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Point-to-Point responses to the Reviewers' comments

Reviewer 1

Concern 1: The literature study is well done and comprehensive. However, there are a number of tools already available on the internet that should be at least listed in the paper. Please check: <https://github.com/topics/bangla-nlp>. I would also appreciate a more detailed description of the text cleaning routines and feature extraction. Perhaps you could include the code in the supporting files section?

Author Response 1: In our endeavor to enrich the literature review section of our manuscript, we initially sought to identify relevant scholarly papers via a specific GitHub link. While this approach yielded only one directly relevant paper concerning our topic of interest, it nonetheless served as a pivotal source of inspiration. Motivated by the findings from this link, we expanded our search to include additional papers that, while not directly sourced from the GitHub repository, were pertinent to the discourse on Bangla fake news detection. This broadened scope allowed us to incorporate a variety of new NLP tools and methodologies into our literature review, significantly enhancing its relevance and depth.

The deliberate inclusion of these papers aligns with our commitment to provide a comprehensive and up-to-date overview of the advancements within the field of NLP, specifically in the context of fake news detection in Bangla. By integrating studies that introduce or employ new NLP tools, we have not only broadened the scope of our literature review but have also highlighted the evolving landscape of research in this area. This approach ensures that our manuscript reflects the current state of technological and methodological developments, thereby offering valuable insights into the potential applications and implications for fake news detection in Bangla.

We believe that this expanded literature review significantly contributes to the academic discourse by delineating the progression of NLP tools and techniques applied to the detection of fake news in Bangla. It underscores the dynamic nature of the field and sets the foundation for our study, positioning it within the broader context of ongoing research efforts. As such, we are confident that our manuscript offers a substantial and timely contribution to the field, providing readers with an in-depth understanding of the current challenges and advancements in Bangla fake news detection.

Author Action 1: We have revised our paper to address the esteemed concern raised on Section II. **RELATED WORKS** and page number 4.

Thank you for bringing this to our attention.

Reviewer 2

Concern 2.1: I think the biggest issue of this draft is the novelty. The proposed RoBERTa-GCN, which is a joint of RoBERTa and GCNs, has been widely discussed previously for text classification.

Author Response 2.1: RoBERTa-GCN for Bangla fake news detection outlines a novel approach that integrates RoBERTa’s advanced language processing capabilities with the Graph Convolutional Network (GCN) model, specifically tailored for the Bangla language. This work highlights the unique linguistic nuances and cultural contexts of Bangla, aiming to address the significant challenge of combating fake news in less commonly represented languages. We’ve trained the model on a diverse dataset, ensuring robust performance against unknown data, and demonstrated its superiority over existing models by achieving impressive accuracy rates. This marks a significant contribution to fake news detection in underrepresented languages, setting a new standard in the field.

Regarding the novelty of the proposed RoBERTa-GCN model, the integration of RoBERTa with GCNs for the specific application of Bangla fake news detection presents several novel aspects:

1. The application of RoBERTa-GCN to the Bangla language, which has been significantly underexplored in the realm of fake news detection, represents a novel contribution. The linguistic and cultural adaptations required for effective fake news detection in Bangla are non-trivial and provide new insights into the application of NLP in less commonly studied languages.
2. Our work involves the creation and utilization of a tailored dataset for Bangla fake news detection, which is an area not thoroughly explored by previous research. The model’s demonstrated performance, surpassing existing models with accuracy rates of 98.60% and 98.75% on benchmark and lab datasets respectively, highlights the effectiveness of our approach.
3. While the individual components (RoBERTa and GCNs) have been previously discussed in the literature, their integration for the purpose of fake news detection, especially within the context of the Bangla language, is innovative. The synergistic use of RoBERTa for deep language understanding and GCNs for capturing relational data in news content is a unique approach tailored for this specific problem.
4. Our work involves the creation and utilization of a tailored dataset for Bangla fake news detection, which is an area not thoroughly explored by previous research. The model’s demonstrated performance, surpassing existing models with accuracy rates of 98.60% and 98.75% on benchmark and lab datasets respectively, highlights the effectiveness of our approach.
5. Focusing on Bangla, a language spoken by a vast population but lacking adequate computational tools for fake news detection, addresses a significant gap in the literature. This regional focus on a problem of global importance, coupled with the development of a high-performing model, contributes new knowledge to the field.

While the combination of RoBERTa and GCNs might have been explored in other contexts, its application to Bangla fake news detection, considering the specific challenges and nuances of the language, presents a novel contribution. Future research can build on this work by exploring other underrepresented languages or extending the model’s capabilities to handle more complex disinformation campaigns.

Author Action 2.1: We have revised our paper to address the esteemed concern raised on Section I INTRODUCTION and page number 2.

Concern 2.2: The presentation of Algo. 1 & 2 should be improved. Such "all text" makes pseudo codes look unprofessional and difficult to understand. It is also unclear to me how to build the "graph" (e.g., how to make a single vector to represent the headline and news content? Is there any pooling process for the RoBERTa embeddings of each word?) during the following GCN.

Author Response 2.2: Thank you for our insightful feedback and specific suggestions, particularly the introduction section. our guidance has helped us to enhance the clarity and relevance of our work. In response to our concerns, we have taken the following actions:

To represent both the headline and the content as a single vector, embeddings from RoBERTa for each word or subword token are first obtained. Then, a pooling operation (e.g., mean pooling or max pooling) is applied across these embeddings. This pooling process reduces the embeddings to a single vector that captures the essence of the entire article, providing a holistic representation that serves as input for the GCN. The graph is constructed by treating the unified representation vector as a feature of a single node or multiple nodes (if segmenting the article further). Edges are defined based on the relationships between these nodes, which could be determined through similarity measures, contextual relationships, or predefined rules capturing the structural and semantic connections within the text.

Author Action 2.2: We have revised our paper to address the esteemed concern raised on Section ROBERTA-GCN and page number 5.

Concern 2.3: The presentation of Fig. 6 should be improved. It is currently not visually appealing and feasible to understand.

Author Response 2.3: After receiving the comment regarding the need for improvement in the presentation of Figure 6, we have undertaken a comprehensive review and enhancement of the image to address the concerns raised. The modifications focused on improving clarity, legibility, visual hierarchy, consistency, simplicity, color scheme, data presentation, and annotation, ensuring that the figure is not only visually appealing but also effectively communicates the intended information.

We have utilized advanced visualization tools and software to refine the figure, making strategic adjustments to font sizes, styles, and the overall color palette to enhance readability and distinguishability. The data visualization elements have been meticulously revised to accurately represent the underlying data, with appropriate scales and intervals. Additionally, we have streamlined the content to focus the viewer's attention on key data points, employing annotations and explanations judiciously to avoid overcrowding while clarifying complex aspects.

We believe these enhancements have significantly improved the visual appeal and understandability of Figure 6. It now aligns better with the document's standards and effectively supports the narrative we aim to convey. Therefore, we consider the comment regarding the improvement of Figure 6 addressed and kindly request the reviewers to consider our efforts in this revised submission.

Author Action 2.3: We have revised our paper to address the esteemed concern raised on subsection IV. THE EXPERIMENTAL ANALYSIS and page number 16.

Concern 2.4: As aimed at Bangla, a more detailed analysis of fake news detection in this specific language is required to make this draft more comprehensive. One suggestion is to have a qualitative comparison (e.g., visualized examples) to demonstrate how the proposed RoBERTa-GCN can perform better than other baselines. Both successful and failed cases should be involved.

Author Response 2.4: In response to the valuable feedback received, we have diligently worked to enrich our manuscript with a detailed exploration of fake news detection in the Bangla language, emphasizing the capabilities and advancements introduced by the RoBERTa-GCN model. We have undertaken the following specific actions to address the concerns raised:

1. We have provided an in-depth examination of the linguistic traits characteristic of Bangla fake news, leveraging examples from our dataset to illustrate these features explicitly. This analysis underlines the complexity of the task and the necessity for a sophisticated model like RoBERTa-GCN.
2. Through visualized examples and case studies, we have showcased the RoBERTa-GCN model's superior performance in accurately classifying fake news articles in Bangla compared to baseline models. This includes a balanced presentation of both successful and failed cases, offering insights into the model's decision-making process and areas of improvement.
3. We supplemented our qualitative analysis with quantitative performance metrics, demonstrating the RoBERTa-GCN model's efficacy through accuracy, precision, recall, and F1 scores. This comparison elucidates the model's significant improvement over existing baseline methods.
4. The manuscript highlights specific adaptations and enhancements made to the RoBERTa-GCN model to cater to the unique challenges presented by the Bangla language, emphasizing its tailored application to this context.

Our revisions aim not only to address the feedback provided but also to significantly contribute to the field of fake news detection in Bangla. By demonstrating the effectiveness of the RoBERTa-GCN model through a blend of qualitative and quantitative analyses, we believe our manuscript now offers a comprehensive and insightful exploration into combating misinformation in less commonly studied languages.

We are confident that these enhancements have thoroughly addressed the comments raised, offering a clearer, more detailed, and comparative perspective on the effectiveness of the RoBERTa-GCN model. We kindly request the reviewers to consider the revisions made and believe that our manuscript now makes a substantial contribution to the field.

Author Action 2.4: We have revised our paper to address the esteemed concern raised on subsection **I INTRODUCTION** and page number **2**.